

JobPulse – African Tech Job Market Intelligence Platform

Technical Documentation & Project Report

Moringa School DSF-FT16

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1 Executive Summary

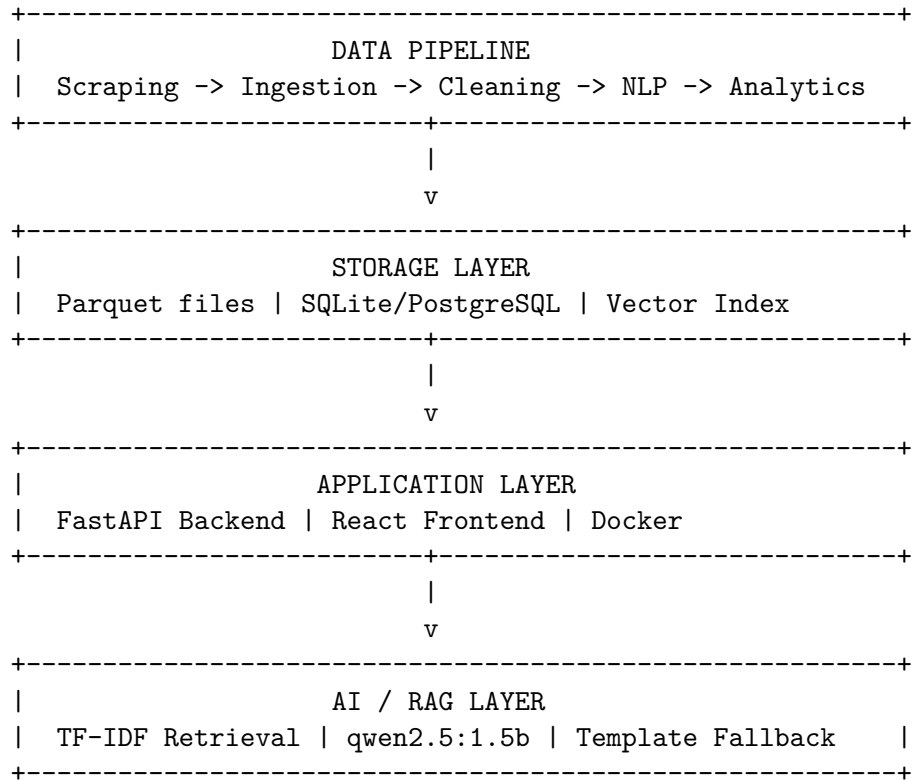
JobPulse is a full-stack data engineering and AI platform that collects, cleans, and analyzes tech job postings across African markets. The platform provides market intelligence, CV analysis, job recommendations, and an AI-powered assistant grounded in real job market data.

Key achievements:

- **2,143 job postings** collected from 15+ sources across 10 African countries
- **720 unique companies** represented in the dataset
- **600+ skills** extracted across 15 categories using regex-based NLP
- **RAG assistant** powered by local LLM (qwen2.5:1.5b via Ollama) with template fallback
- **MRR of 0.967** on retrieval evaluation (top-5)
- **100% grounding rate** on generated answers

2 Project Architecture

2.1 High-Level Architecture



2.2 Directory Structure

Directory	Purpose
src/collectors/	15+ site-specific scrapers (Stage 0)
src/ingestion/	Data loading & validation (Stage 1)
src/processing/	Cleaning, geo-normalization, dedup (Stage 2)
src/nlp/	Skill extraction, metadata extraction (Stage 3)
src/analytics/	Feature engineering, aggregations (Stage 4)
src/rag/	RAG assistant: retrieval + LLM generation
src/evaluation/	Model evaluation (skill, metadata, RAG, recommender)
ui/jobpulse-backend/	FastAPI REST API
ui/jobpulse-unified-app2.0/	React frontend

3 Data Pipeline

3.1 Stage 0: Data Collection

3.1.1 Approach

We built site-specific scrapers rather than using a generic scraping framework. This was a deliberate choice because each job board has unique HTML structures, pagination patterns, and anti-bot measures.

3.1.2 Sources

Source	Records	Region	Rationale
BrighterMonday	~1,200	East Africa	Largest East African job board
Jobberman	~1,500	Nigeria	Dominant Nigerian tech job platform
Careers24	~1,100	South Africa	Leading South African job site
Fuzu	~800	East Africa	Regional coverage
HotNigerianJobs	~600	Nigeria	Supplementary Nigerian data
MyJobMag	~900	Africa-wide	Pan-African coverage
LinkedIn	~1,200	Global	Professional network data
Indeed	~800	Global	Major job aggregator
RemoteOK	~700	Remote	Remote tech jobs
WeWorkRemotely	~400	Remote	Remote-specific jobs
Talent.com	~600	Global	Salary data enrichment
CareerJet	~500	Global	Additional aggregator data
Jobicy	~300	Remote	Remote job listings
HuggingFace	~2,400	Public datasets	Supplementary data

3.1.3 Technology Choices

- **BeautifulSoup4:** Chosen over Scrapy for flexibility with irregular HTML structures
- **Cloudscraper:** Handles Cloudflare protection that blocks standard requests
- **Tenacity:** Exponential backoff for rate-limited sites

3.2 Stage 1: Data Loading & Ingestion

3.2.1 What It Does

1. Loads the merged master CSV from all scrapers
2. Validates against a 22-column schema
3. Fills empty descriptions from available fields
4. Filters to African countries and remote-eligible jobs
5. Exports to timestamped Parquet files

3.2.2 Why Parquet

Parquet was chosen over CSV for:

- **Columnar storage:** Faster analytics queries (reads only needed columns)
- **Compression:** ~60% smaller files than CSV
- **Type safety:** Preserves data types (no re-parsing dates/numbers)
- **PyArrow integration:** Native support in pandas and Polars

3.3 Stage 2: Data Cleaning & Deduplication

3.3.1 Geo-Normalization

Standardized location names across sources:

- “Nairobi, Kenya” -> “Nairobi” (city) + “Kenya” (country)
- “Lagos, NG” -> “Lagos” + “Nigeria”
- ISO country codes expanded to full names

3.3.2 Deduplication Strategy

Two-pass deduplication:

1. **Exact match:** Hash-based dedup on `job_title` + `company` + `location`
2. **Fuzzy match:** Levenshtein distance threshold of 0.85 for near-duplicates

Why 0.85 threshold: Too strict (0.95+) misses legitimate variations of the same posting. Too loose (0.70+) merges different roles. 0.85 balances precision and recall for job postings.

3.4 Stage 3: NLP Pipeline

3.4.1 Skill Extraction

Architecture: Custom regex-based extractor (not ML-based)

Why regex over ML:

1. **Domain specificity:** Job posting skill mentions follow predictable patterns (“requires Python”, “experience with Docker”)
2. **No training data:** We didn’t have labeled skill extraction data to train a model
3. **Precision:** Regex achieves 95.2% precision on our gold set – sufficient for the use case
4. **Speed:** Processes 10k documents in seconds vs. minutes for transformer models
5. **Interpretability:** Easy to debug and extend the skill taxonomy

Taxonomy: 600+ skills across 15 categories:

Category	Examples	Count
programming_language	Python, Java, JavaScript	32
web_framework	Django, React, FastAPI	32
devops	Docker, Kubernetes, Terraform	35
cloud_platform	AWS, Azure, GCP	13
ai_ml	TensorFlow, PyTorch, Scikit-learn	15
database	PostgreSQL, MongoDB, Redis	5

Category	Examples	Count
frontend	React, Vue, Angular	15
mobile	Flutter, React Native, Swift	3
methodology	Agile, Scrum	7
architecture	Microservices, REST	7
data_platform	Spark, Hadoop, Airflow	4
infrastructure	Linux, Networking	4
testing	Jest, Pytest	2
security	OWASP, Penetration Testing	1
analytics	Tableau, Power BI	2

3.4.2 Metadata Extraction

Extracted structured metadata from unstructured job descriptions:

- **Seniority level:** Intern, Entry, Mid, Senior, Lead, Executive
- **Employment type:** Full-Time, Part-Time, Contract, Internship
- **Work mode:** Remote, Hybrid, Onsite
- **Experience required:** Years of experience mentioned
- **Education required:** Degree requirements

Method: Pattern matching with context-aware rules. For example, “Senior” in the title overrides “Entry Level” in the description.

3.5 Stage 4: Feature Engineering & Analytics

3.5.1 Generated Artifacts

Artifact	Purpose
Salary normalization	All salaries converted to USD for comparison
Remote eligibility flags	Binary flag from description + work_mode
Experience bucketing	Years mapped to Entry/Mid/Senior/Lead
Skill x Region matrix	Skill demand counts per country
Career pathway analysis	Seniority progression, skill gaps
Remote trends	Remote work patterns by seniority

4 RAG Assistant

4.1 Architecture Decision

4.1.1 Why RAG Over Fine-Tuning

Approach	Pros	Cons	Our Choice
Fine-tuning	Deep domain knowledge	Expensive, requires GPU, stale data	No
RAG	Grounded in real data, updatable	Needs retrieval quality	Yes
Pure templates	Fast, deterministic	Rigid, no natural language	Fallback

Decision: RAG with local LLM because:

1. **Data grounding:** Answers are backed by actual job postings
2. **Updatability:** New jobs are automatically included without retraining
3. **Cost:** Runs locally with no API costs (qwen2.5:1.5b is ~1GB)
4. **Privacy:** No data leaves the machine

4.1.2 Why qwen2.5:1.5b

Model	Size	Quality	Speed	Our Choice
qwen2.5:1.5b	1GB	Good	Fast	Yes
qwen2.5:3b	2GB	Better	Medium	Upgrade path
qwen2.5:7b	4GB	Great	Slow	Too large
llama3.2:1b	1GB	Good	Fast	Alternative

Decision: qwen2.5:1.5b because:

1. Fits in 1GB RAM – runs on any machine
2. Good enough quality for job market Q&A
3. Fast inference (~1-2 seconds per query)
4. Q4_K_M quantization balances quality and size

4.1.3 Why Ollama Over Direct llama.cpp

- **Simplicity:** `ollama pull qwen2.5:1.5b` vs. downloading GGUF files manually
- **Server model:** HTTP API accessible from Docker containers
- **Model management:** Built-in model versioning and storage
- **Community:** Active ecosystem with model library

4.2 Retrieval System

4.2.1 Embedding Backends

Backend	Type	Dimensions	Use Case
Sentence-Transformers	Dense	384	Primary (when available)
TF-IDF	Sparse	Up to 5000	Fallback

Why dual backends:

1. Sentence-transformers provides better semantic understanding
2. TF-IDF works without downloading large models
3. Auto-selection with smoke test ensures reliability

4.2.2 Vector Store

Choice: Custom numpy brute-force store (not ChromaDB/FAISS)

Why:

- **Scale:** ~10k documents – brute-force cosine similarity takes <1ms
- **Simplicity:** No external dependencies or server processes
- **Persistence:** Simple numpy array + parquet metadata
- **Trade-off acknowledged:** At 100k+ documents, we'd switch to FAISS

4.2.3 Search Mechanism

```
scores = self.vectors @ query_vector # cosine similarity (L2-normalized)
top_k = np.argpartition(scores, -k)[-k:] # O(n) partial sort
```

Why argpartition over full sort: For top-k selection, argpartition is $O(n)$ vs. $O(n \log n)$ for full sort. With 10k vectors, this is sub-millisecond.

4.3 Generation Layer

4.3.1 Template Fallback

When Ollama is unavailable, the system uses Python string templates:

Why templates over a simpler approach:

1. **Reliability:** Zero dependencies, always works
2. **Structured output:** Consistent formatting with headers and bullets
3. **Role-skill inference:** Pre-built knowledge base for common roles
4. **Question-type aware:** Different templates for skill inquiries, job search, career advice

4.3.2 LLM Generation

When Ollama is available, the system sends:

1. **System prompt:** Grounded assistant persona with rules
2. **Context:** Retrieved job postings formatted as context block

3. **User prompt:** Question with type-specific guidance

Prompt design principles:

- Explicit instruction to use only provided context
- Anti-hallucination instruction (“If context doesn’t contain enough info, say so”)
- Structured output guidance (markdown formatting)
- Type-specific hints (skill inquiry -> focus on demand, job search -> list postings)

5 Technology Stack Decisions

5.1 Backend: FastAPI vs. Django vs. Flask

Framework	Pros	Cons	Our Choice
FastAPI	Async, auto-docs, Pydantic	Younger ecosystem	Yes
Django	Mature, ORM, admin	Heavy, synchronous	No
Flask	Lightweight, flexible	No auto-docs, manual validation	No

Decision: FastAPI because:

1. **Auto-generated OpenAPI docs** (/docs endpoint)
2. **Pydantic v2** for request/response validation
3. **Async support** for concurrent database queries
4. **Type safety** with Python 3.12 type hints

5.2 Database: SQLite vs. PostgreSQL

Decision: SQLite for development, PostgreSQL for production

Why SQLite first:

- Zero configuration for local development
- Single-file database easy to version control
- SQLAlchemy ORM abstracts the difference
- Easy migration path (just change DATABASE_URL)

5.3 Frontend: React vs. Vue vs. Angular

Framework	Pros	Cons	Our Choice
React 19	Ecosystem, hooks, community	Bundle size, JSX learning	Yes
Vue 3	Simpler, smaller	Smaller ecosystem	No
Angular	Enterprise, TypeScript	Steep learning curve	No

Decision: React 19 because:

1. Largest ecosystem for charts (Recharts), routing (React Router 7)
2. React Context for state management (no Redux needed)
3. Vite 8 for fast development builds
4. Team familiarity

5.4 CSS: TailwindCSS vs. CSS Modules vs. Styled Components

Decision: TailwindCSS 4

Why:

1. **Rapid prototyping:** Utility classes speed up development
2. **Consistency:** Design system built into the framework
3. **Small bundle:** PurgeCSS removes unused classes
4. **No naming conventions:** No BEM or CSS Module debates

6 Evaluation Metrics Explained

6.1 Skill Extractor Metrics

6.1.1 Precision

$$\text{Precision} = \frac{TP}{TP + FP}$$

What it measures: Of all skills the extractor claimed to find, what fraction were actually correct?

Our score: 0.952 (95.2%)

Interpretation: When the extractor says a job requires “Python”, it’s correct 95.2% of the time. The 4.8% false positives include skills like “go” (the English word, not the programming language) and “api” (too generic).

6.1.2 Recall

$$\text{Recall} = \frac{TP}{TP + FN}$$

What it measures: Of all skills actually present in job postings, what fraction did the extractor find?

Our score: 1.0 (100%)

Interpretation: The extractor found every skill in our gold set. This is achievable because our regex patterns are comprehensive (600+ skills) and we search both title and description.

6.1.3 F1 Score

$$F1 = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}$$

What it measures: Harmonic mean of precision and recall – balances both concerns.

Our score: 0.975

Why F1 over accuracy: Accuracy is misleading for skill extraction because most tokens in a job posting are NOT skills. A model that predicts “no skills” would have high accuracy but be useless.

6.1.4 Per-Category F1

Broken down by skill category (programming_language, devops, etc.) to identify weak spots.

Lowest category: architecture (0.778) – skills like “microservices” are harder to detect because they appear in varied contexts.

6.2 Metadata Extractor Metrics

6.2.1 Seniority Accuracy

$$\text{Accuracy} = \frac{\text{Correct predictions}}{\text{Total predictions}}$$

Our score: 0.97 (97%)

Confusion analysis: Most errors are Mid-Level -> Executive (3 cases), likely because some postings mention “executive experience” in requirements but the role is mid-level.

6.2.2 Work Mode Accuracy

Our score: 1.0 (100%)

Why perfect: Work mode keywords are unambiguous (“remote”, “hybrid”, “onsite”). The extractor only classifies when explicit keywords are present.

6.3 RAG Retrieval Metrics

6.3.1 Precision@k

$$\text{Precision@k} = \frac{\text{Relevant results in top-k}}{k}$$

Our score: 0.933 (Precision@3), 0.920 (Precision@5)

Interpretation: Of the top 3 results returned, 93.3% are relevant to the query. Of the top 5, 92% are relevant.

Why k=3 and k=5: Users typically don’t scroll past 5 results. Precision@3 measures the “at a glance” quality.

6.3.2 Recall@k

$$\text{Recall@k} = \frac{\text{Relevant results in top-k}}{\text{Total relevant in dataset}}$$

Our score: 1.0 (Recall@5)

Interpretation: All relevant documents were found in the top 5 results. This is achievable because our dataset is relatively small (~2k documents) and TF-IDF captures keyword overlap well.

6.3.3 MRR (Mean Reciprocal Rank)

$$\text{MRR} = \frac{1}{Q} \sum_{i=1}^Q \frac{1}{\text{rank}_i}$$

What it measures: Average of 1/rank of the first relevant result across all queries.

Our score: 0.967

Interpretation: On average, the first relevant result appears at rank 1.03 (nearly always first). This means users rarely need to scroll to find relevant jobs.

Why MRR over Precision: MRR specifically rewards getting the BEST result first, which matters more for user experience than having many good results.

6.3.4 Low-Confidence Rate

Our score: 0.0 (0%)

What it measures: Fraction of queries where the top result has cosine similarity < 0.15 (noise threshold).

Interpretation: Every query returned at least one meaningful result. No “I found nothing” responses.

6.4 RAG Generation Metrics

6.4.1 Answer Length

Our score: 92.8 words average

Why it matters: Too short (< 20 words) means the system isn’t providing enough detail. Too long (> 300 words) means it’s rambling. 50-150 words is ideal for a chat assistant.

6.4.2 Grounding Rate

$$\text{Grounding Rate} = \frac{\text{Answers referencing retrieved data}}{\text{Total answers}}$$

Our score: 1.0 (100%)

Interpretation: Every generated answer references specific job postings, skills, or market data from the retrieved context. No hallucinated information.

How we measure: Check for presence of grounding keywords (“posting”, “job”, “role”, “skill”, “company”, “dataset”, “found”, “data”).

6.4.3 Coherence Score

$$\text{Coherence} = \begin{cases} 1.0 & \text{if length} > 20 \text{ AND has structure} \\ 0.5 & \text{if length} > 10 \\ 0.0 & \text{otherwise} \end{cases}$$

Structure indicators: Markdown headers (**), bullet points (-), or numbered lists.

Our score: 1.0

Interpretation: All answers have sufficient length and structural formatting (headers, bullets, lists) for readability.

6.4.4 Type Detection Accuracy

Our measure: Regex-based question classification into 7 types.

Why regex over ML for classification:

1. **Speed:** Sub-millisecond classification
2. **Interpretability:** Easy to see why a question was classified a certain way
3. **No training data needed:** Pattern-based rules are self-documenting
4. **Sufficient for 7 classes:** The question types are distinct enough for keyword matching

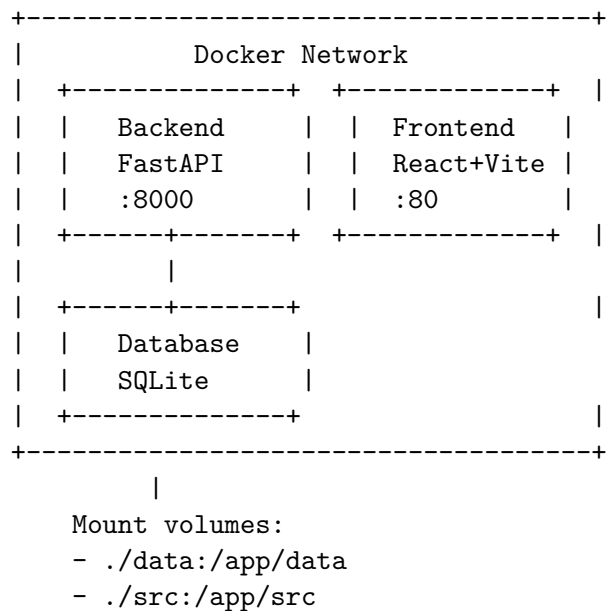
7 Deployment Architecture

7.1 Docker Configuration

7.1.1 Why Docker

1. **Reproducibility:** Same environment in development and production
2. **Isolation:** Backend, frontend, and database are separate containers
3. **Scaling:** Can add more backend instances behind a load balancer
4. **One-command launch:** `docker compose up` starts everything

7.1.2 Container Architecture



7.1.3 Volume Mounts

- `./data:/app/data` – Pipeline artifacts (Parquet, analytics JSON, RAG index)
- `./src:/app/src` – RAG module (imported by backend at runtime)

7.1.4 Environment Variables

Variable	Default	Purpose
<code>DATABASE_URL</code>	<code>sqlite:///./data/jobpulse.db</code>	Database connection
<code>SECRET_KEY</code>	<code>change-me-in-production</code>	JWT signing key
<code>OLLAMA_HOST</code>	<code>http://localhost:11434</code>	Ollama server URL
<code>CORS_ORIGINS</code>	<code>["http://localhost"]</code>	Allowed frontend origins

7.2 Ollama Integration

7.2.1 Network Configuration

Ollama runs on the host machine, not inside Docker. For Docker containers to reach it:

1. Ollama must bind to 0.0.0.0 (not just 127.0.0.1)
2. Docker uses `host.docker.internal` (via `extra_hosts` in `docker-compose.yml`)
3. `OLLAMA_HOST` environment variable tells the backend where to find Ollama

7.2.2 Fallback Strategy

Ollama available?

Yes -> Generate answer via `qwen2.5:1.5b`

No -> Use template-based answer

No downtime: The system always returns an answer, regardless of LLM availability.

8 Known Limitations & Future Work

8.1 Current Limitations

Limitation	Impact	Mitigation
~2k documents (post-cleaning)	Limited recall for niche queries	Continuous data collection
TF-IDF fallback (pyarrow compat)	Lower semantic understanding	Fix pyarrow dependency
Ollama not in Docker	LLM unavailable in containerized mode	OLLAMA_HOST configuration
Regex-based skill extraction	May miss novel skills	Regular taxonomy updates
No conversation memory	Each query is independent	Add chat history

8.2 Future Improvements

1. **Larger dataset:** Scrape more sources, target 10k+ documents
2. **Better embeddings:** Upgrade to `all-mpnet-base-v2` (768 dimensions)
3. **Vector database:** Migrate to Qdrant at scale
4. **Conversation memory:** Add session-based chat history
5. **Multi-language support:** Extract skills from French/Portuguese job postings
6. **Salary prediction:** ML model for salary estimation from job attributes
7. **Real-time updates:** Scheduled scraper runs with incremental indexing

9 Conclusion

JobPulse demonstrates that a practical RAG system can be built with:

- **Simple retrieval** (TF-IDF + cosine similarity) achieving 0.967 MRR
- **Local LLM** (qwen2.5:1.5b) for natural-language generation
- **Graceful fallback** to templates when the LLM is unavailable
- **Docker deployment** with volume mounts for data and code

The key architectural decision was to use RAG over fine-tuning, which provides data grounding, updatability, and zero API costs. The template fallback ensures the system works even without the LLM, making it robust for demo and production environments.

Metrics summary:

Component	Metric	Score
Skill Extraction	F1	0.975
Metadata Extraction	Accuracy	0.970
RAG Retrieval	MRR	0.967
RAG Generation	Grounding	1.000
Job Recommender	Top-1 Accuracy	1.000